Electrical Safety and Lockout/Tagout (LOTO) Procedures

Reference: 29 CFR 1910.147 / 1910.301-399

Electrical Hazard Overview

Electrical hazards cause over 300 deaths and 4,000 injuries annually in the U.S. Primary electrical hazards: electric shock (as low as 10mA can cause involuntary muscle reaction, 100mA can be fatal), arc flash (temperatures up to 35,000 degreesF), arc blast (pressure waves up to 2000 lb/ft2), and fire from electrical faults.

Lockout/Tagout (LOTO) - 29 CFR 1910.147

LOTO prevents unexpected energization of equipment during servicing. Required steps: 1-Notify affected employees, 2-Identify all energy sources, 3-Shut down equipment, 4-Isolate energy sources, 5-Apply lockout/tagout devices, 6-Release stored energy (bleed, discharge, block), 7-Verify zero energy state before starting work.

Energy Isolation Devices

Acceptable lockout devices: padlocks (individually keyed, one key per worker), hasps (allows multiple locks for group lockout), lockout blocks, valve covers, and blanking flanges. Tags alone are NOT adequate when locks can be applied. Each authorized employee must apply their own personal lock.

Arc Flash Safety

Arc flash boundary categories require appropriate PPE: Incident Energy 1.2-4 cal/cm2 requires Arc-Rated clothing 4 cal/cm2 minimum. 4-8 cal/cm2 requires AR clothing 8 cal/cm2. 8-25 cal/cm2 requires AR clothing 25 cal/cm2. Over 25 cal/cm2 requires 40 cal/cm2 AR suit. Always verify voltage before work with calibrated meters.

Ground Fault Circuit Interrupter (GFCI)

GFCIs are required for all 120V single-phase 15A/20A receptacles in construction, near water, and in temporary installations. GFCIs detect current imbalance of 4-6mA and trip in 1/40 second. Test GFCIs monthly using the Test/Reset button. Replace any GFCI that fails testing immediately.

Electrical PPE Requirements

When working on or near energized electrical equipment: insulated gloves rated for voltage, arc flash rated face shield and balaclava, arc-rated clothing, insulated tools, rubber mats, and safety glasses. Never work alone on energized systems. Minimum approach distances must be maintained per NFPA 70E.